

DBMS Lab Task - Library Management System

Task Title: Library Management System - Query Practice

Description:

Create and query a small Library Management Database to manage books, members, and issued records.

You will design tables, insert data, and write SQL queries to perform various operations.

Table Structure:

1. Books

book_id INT PRIMARY KEY,
title VARCHAR(100),
author VARCHAR(100),
category VARCHAR(50),
price DECIMAL(8,2),
available_copies INT

2. Members

member_id INT PRIMARY KEY,
name VARCHAR(100),
membership_date DATE,
email VARCHAR(100)

3. IssueRecords

issue_id INT PRIMARY KEY,
book_id INT REFERENCES Books(book_id),
member_id INT REFERENCES Members(member_id),
issue_date DATE,

return_date DATE

Queries to Perform:

1. List all books along with their authors and categories.
2. Display member names who have borrowed at least one book.
3. Find the total number of books issued per member.
4. Show the books that are not currently available (available_copies = 0).
5. Display members who joined before the year 2022.
6. Calculate the total revenue if each book's price is multiplied by the number of copies issued.
7. Show all books that belong to the category 'Technology'.
8. Find members who have not returned books yet (return_date IS NULL).
9. Display the most recently issued book (latest issue_date).
10. Calculate the average book price per category.